Project Report

Auto Ticket Classification using Flow Designer

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Project Report Format

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1. INTRODUCTION

1.1 Project Overview

Organizations receive a large number of support tickets every day through email, portals and chat. Sorting these tickets by hand into the right category and assigning them to the right support group is slow and often inconsistent. This project, **Auto Ticket Classification using Flow Designer**, automates that process on the ServiceNow platform. A flow built in Flow Designer is triggered whenever a new incident is created. It reads the short description of the ticket, matches it against defined keywords, sets the correct category, and assigns the ticket to the appropriate support group without any manual effort.

The solution uses a no-code/low-code approach. Flow Designer provides a visual interface where triggers, conditions and actions are combined into a workflow, so the logic can be created, tested and modified easily by administrators.

1.2 Purpose

The purpose of this project is to reduce manual effort in ticket triage and to make sure every ticket reaches the right team quickly. The main objectives are:

- Automatically classify incoming tickets into categories such as Network, Software, Hardware, Database and Access.
- Automatically assign each ticket to the correct assignment group.
- Reduce response and resolution time by removing manual triage.
- Improve accuracy and consistency of ticket categorization.
- Demonstrate the use of Flow Designer for practical workflow automation.

2. IDEATION PHASE

2.1 Problem Statement

Support desk agents spend a considerable amount of time reading each new ticket and deciding its category and owner. Human error, differing interpretations and heavy ticket volume lead to wrongly categorized tickets, repeated re-assignments, delayed resolutions and dissatisfied users. There is a need for an automated, rule-based mechanism that classifies tickets the moment they are created.

I am	I am trying to	But	Because	Which makes me feel
A support desk agent / IT administrator	Route every new ticket to the right team quickly	Tickets are often wrongly categorized and assigned	Classification is done manually and depends on individual judgement	Overloaded, delayed and unable to meet SLA targets

2.2 Empathy Map Canvas

Section	Details
Says	"Why was my ticket sent to the wrong team?" / "I need this fixed quickly."
Thinks	Manual sorting wastes time. Automation would let the team focus on solving issues.
Does	Reads each ticket, selects a category, assigns a group, and re-assigns when a mistake is found.
Feels	Frustrated by repetitive work, stressed by SLA pressure, worried about errors.
Pain	Delays, mis-routing, inconsistent categories, high ticket volume.
Gain	Instant classification, correct routing, faster resolution, happier users.

2.3 Brainstorming

The team discussed different ways to solve the problem and compared them:

Idea	Description	Decision
Manual triage by agents	Agents read and classify every ticket	Rejected - slow and error-prone
Business rules with scripts	Server-side scripts to set category on insert	Considered - needs coding and is harder to maintain
Flow Designer automation	Visual flow triggered on incident creation with keyword conditions	**Selected** - low-code, easy to maintain and extend
ML-based prediction	Train a classifier on historical tickets	Kept for future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	User Action	System Action	Outcome
Raise ticket	User submits an incident with a short description	Incident record is created	Ticket logged
Trigger	-	Flow Designer flow starts on record creation	Automation begins
Classify	-	Flow checks keywords in the description and sets the category	Category assigned
Assign	-	Flow sets the assignment group based on the category	Ticket routed
Resolve	Support team works on the ticket	Ticket is updated and closed	User issue resolved

3.2 Solution Requirement

Functional Requirements

- Trigger the flow automatically when a new incident is created.

- Read the short description / description of the ticket.
- Set the category according to matching keywords.
- Assign the ticket to the appropriate assignment group.
- Handle tickets that match no rule by assigning them to a default group.

Non-Functional Requirements

- Classification should complete within seconds of ticket creation.
- The flow should be easy to modify without coding.
- The solution should be reliable, secure and role-based.
- The design should scale as ticket volume grows.

3.3 Data Flow Diagram

The data flow of the system is described below (Level 0 to Level 1):

Step	Process	Data
1	User creates a ticket	Short description, caller, priority
2	Incident table stores the record	Incident record
3	Flow Designer trigger fires (Created)	Incident sys_id
4	Condition checks match keywords	Description text
5	Update Record action sets category and assignment group	Category, Assignment group
6	Ticket appears in the queue of the assigned team	Updated incident

3.4 Technology Stack

Component	Technology
Platform	ServiceNow (Personal Developer Instance)
Automation tool	Flow Designer
Data store	Incident table (ServiceNow database)
Trigger	Record created on Incident table
Actions	Update Record, Conditions / If-Else logic
Access	Web browser

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution Fit
Manual and slow ticket classification	Flow classifies each ticket instantly on creation

Problem	Solution Fit
Tickets sent to the wrong team	Keyword-based rules route tickets to the correct group
Inconsistent categorization	One standard rule set applied to every ticket
Need for coding to automate	Flow Designer offers a visual, low-code approach

4.2 Proposed Solution

A flow named **Auto Ticket Classification** is created in Flow Designer. Its trigger is *Created* on the Incident table. Inside the flow, condition blocks check whether the short description contains keywords for each category, for example network, VPN or wifi for Network; software, install or application for Software; laptop, printer or keyboard for Hardware; and password, login or access for Access. When a condition matches, an *Update Record* action sets the Category and Assignment Group on the incident. A default branch handles tickets that do not match any keyword.

4.3 Solution Architecture

Layer	Description
Input layer	User raises an incident through the portal, email or form
Trigger layer	Flow Designer trigger: Incident Created
Logic layer	If / Else conditions matching keywords to categories
Action layer	Update Record: set Category and Assignment Group
Output layer	Classified and assigned ticket in the support team queue

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activity	Duration
Phase 1	Requirement gathering and problem understanding	Week 1
Phase 2	Ideation, empathy map and brainstorming	Week 1
Phase 3	Instance setup, creating assignment groups and categories	Week 2
Phase 4	Building the flow in Flow Designer	Week 2 - 3
Phase 5	Testing with sample tickets and fixing issues	Week 3
Phase 6	Documentation and final report	Week 4

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The flow was tested by creating sample incidents with different descriptions and checking the category and assignment group set by the flow.

Test Case	Sample Short Description	Expected Result	Status
TC-01	VPN is not connecting	Category: Network, Network team	Pass
TC-02	Unable to install application	Category: Software, Software team	Pass
TC-03	Laptop keyboard not working	Category: Hardware, Hardware team	Pass
TC-04	Forgot my password	Category: Access, Service Desk	Pass
TC-05	General enquiry	Default group assigned	Pass

The flow executed within a few seconds of ticket creation for every test case, and the execution details in Flow Designer showed all actions completed successfully.

7. RESULTS

7.1 Output Screenshots

The following screenshots show the flow design, execution details and classified incidents.

[Screenshot 1: Flow Designer - flow trigger and actions]

[Screenshot 2: Flow execution details]

[Screenshot 3: Incident with auto-assigned category and assignment group]

8. ADVANTAGES & DISADVANTAGES

Advantages

- Removes manual effort in ticket triage.
- Faster routing and quicker resolution.
- Consistent and accurate categorization.
- Low-code: easy to build, change and maintain.
- Scalable to handle a growing number of tickets.

Disadvantages

- Keyword-based rules may miss unusual wording.
- Rules must be updated when new categories or terms appear.
- Does not understand context the way an ML model can.

9. CONCLUSION

The Auto Ticket Classification project shows that Flow Designer can automate ticket categorization and assignment effectively. By triggering a flow on incident creation and applying keyword-based rules, tickets are classified and routed to the right team immediately. This reduces manual work, shortens response time and improves consistency, making the support process more efficient.

10. FUTURE SCOPE

- Integrate machine learning (Predictive Intelligence) for smarter classification.
- Add priority prediction and automatic SLA assignment.
- Send notifications to users and assignees through email or chat.
- Support multiple languages in ticket descriptions.
- Extend the flow to requests, problems and change tickets.

11. APPENDIX

Source Code (if any)

The solution is built using Flow Designer (no-code). No custom source code is required. Flow configuration is available in the ServiceNow instance.

Dataset Link

Not applicable - sample incidents were created directly in the ServiceNow instance for testing.

GitHub & Project Demo Link

GitHub: [Add link here]
Project Demo: [Add link here]